

Lecture 1: Introduction to Time Series Econometrics

Nicky L. Grant

September 2016

Table of contents

1. Course Details
2. Introduction to Time Series
3. Sample Realisations of Time Series Processes
4. Quick Stats Recap
5. Key Definitions
6. Theoretical Properties of MA Process
7. Recap
8. Next Week

Course Details

Some Background Information

Time Series Econometrics- tool to analyse economic time series data

Particularly macroeconomic and financial data

Builds on time series methods in **ECON20110 Basic Econometrics**

*Prerequisite for **ECON31012 Financial Econometrics***

Mixture of theoretical & empirical methods

Will use EViews software

Nicky Grant (wk. 1-4) - **Prof. Alastair Hall** (wk 6-10)

Nicky: Mon 2-3pm & Fri 1-2pm (ALB 2.007)

Alastair: Tues: 3-4pm & Fri 1-2pm.

Course roughly split in to 6 parts over 8 lectures.

1. Introduction to Time Series
 2. Stationarity and stationary processes: (AR , MA , $ARMA$)
 3. Estimation, Hypothesis Testing & Model Selection
 4. Prediction & Seasonality
-
5. System modelling: Vector autoregressive (VAR) models
(*Extends 1-4 for multivariate models (not inc. seasonality)*)
 6. Random Walks & Unit roots.

Slides available on Blackboard in advance of each lecture

Lecture notes covering the topics in lecture

- Text books also help understanding
- Enders *Applied Econometric Time Series* recommended

Classes start week 3 (PC Labs and exercises).

- Attempt exercise questions before class (**Extremely Important**)
- Including applied work on EViews, using real data (provided)

Assessment:

20% empirical project

Two weeks to complete- submit last week of term.

Will use EViews to analyse data

80% final examination in January

1.5 hours duration

2 support sessions in last weeks of term (2 hours each)

Not compulsory

Mock & past exam papers on Blackboard before Christmas Break.

Introduction to Time Series

How to infer properties of a variable(s) observed over time?

e.g Interest Rates, Unemployment, Stock Prices.

Model properties of time series **esp. dependence over time**

future interest rates on past interest rates (univariate)

future inflation on past interest rates (multivariate)

Differences with Cross Section:

Temporal ordering

Potential dependence between observations

Time Series Process

Time Series Process Y_t ($t = \dots, -1, 0, 1, \dots$)

[Y_t is a Random Variable]

Observe a **Realisation** $\{y_t\}_{t=1}^T$

[Data Sample of size T]



Sample realisations from $Y_t = 0.5 + 0.4Y_{t-1} + \varepsilon_t$ where $\varepsilon_t \sim N(0, 1)$

Aim: Develop methods to infer properties of the true **Process** using **Sample Realisation**.

Time Series vs. Sample Realisation: Empirical Example

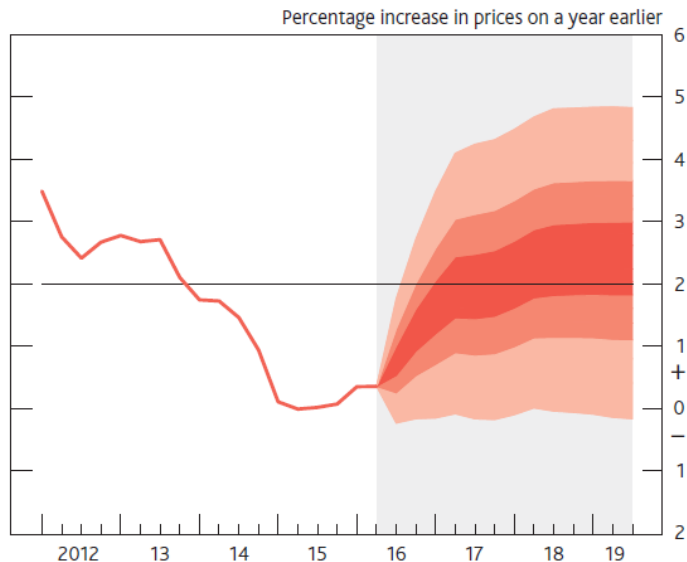


Figure 1: UK CPI and Projections Based on Market Expectations in August 2016

Outturns of inflation are expected to lie within each pair of the lighter red areas on 30 occasions. In any particular quarter of the forecast period, inflation is therefore expected to lie somewhere within the fans on 90 out of 100 occasions. On the remaining 10 out of 100 occasions inflation can fall anywhere outside the red area

SOURCE: BANK OF ENGLAND INFLATION REPORT, AUGUST 2016, PG. 39

Sample Realisations of Time Series Processes

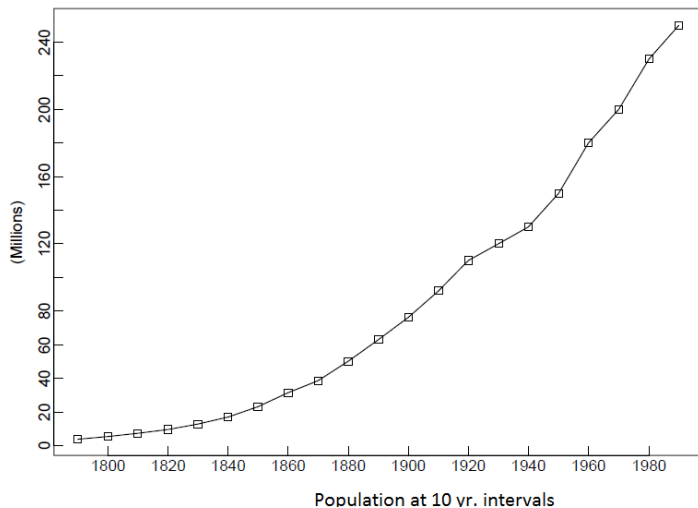


Figure 2: US Population at Ten Year Intervals

SOURCE: INTRODUCTION TO TIME SERIES (BROCKWELL), PG. 5

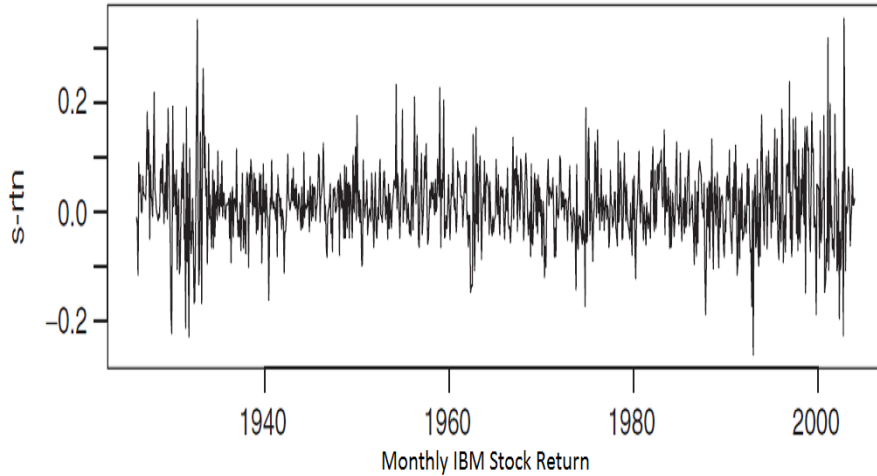


Figure 3: Monthly IBM Stock Return

SOURCE: ANALYSIS OF FINANCIAL TIME SERIES (TSAY), PG.18

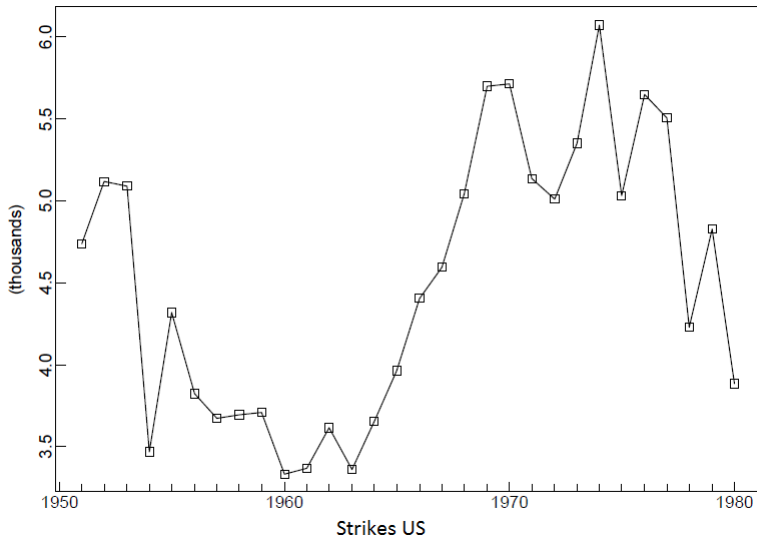


Figure 4: Total Yearly US Strikes

SOURCE: INTRODUCTION TO TIME SERIES (BROCKWELL), PG.5

Quick Stats Recap

Stats Recap: Moments of Distributions

Definition: Expectation Operator

For any r.v.'s X, Y and any function $g(\cdot, \cdot)$ the operator $\mathbb{E}[\cdot]$ satisfies

$$\mathbb{E}[g(X, Y)] := \sum_x \sum_y g(x, y) \Pr\{X = x, Y = y\} \quad \text{DISCRETE}$$

$$\mathbb{E}[g(X, Y)] := \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) dx dy \quad \text{CONTINUOUS}$$

Example: Moments of a Random Variable

$$\mathbb{E}[X]$$

Mean of X

$$\mathbb{E}[(Y - E[Y])^2]$$

Variance of Y

$$\mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

Covariance of X, Y

Key Definitions

Stationarity

Definition: **Stationary Process**

A process Y_t is **Stationary** if

1. $\mathbb{E}[Y_t] = \mu$ for all t
2. $\text{Var}[Y_t] = \sigma^2 < \infty$ for all t
3. $\text{Cov}[Y_t, Y_{t-j}] = \gamma(|j|)$ for each j & for all t

Time Series Process has desirable properties if is Stationary

The assumption may not hold in practise

1. Variance of Y_t changes over time
2. Mean of Y_t changes over time
3. Y_t has a unit root (e.g a Random Walk) [Lecture 8]

Definition: **White Noise**

A process Y_t is **White Noise** if

1. $\mathbb{E}[Y_t] = 0$ for all t
2. $\mathbb{E}[Y_t^2] = \sigma^2 < \infty$ for all t
3. $\mathbb{E}[Y_t Y_{t-j}] = 0$ for all t & for any $j \neq 0$

Shorthand notation $Y_t \sim \text{WN}(\sigma^2)$

Note that: White Noise $\Rightarrow$ Stationarity, but not the reverse

Some Simple Univariate Time Series Models

Autoregressive: depends on own past values

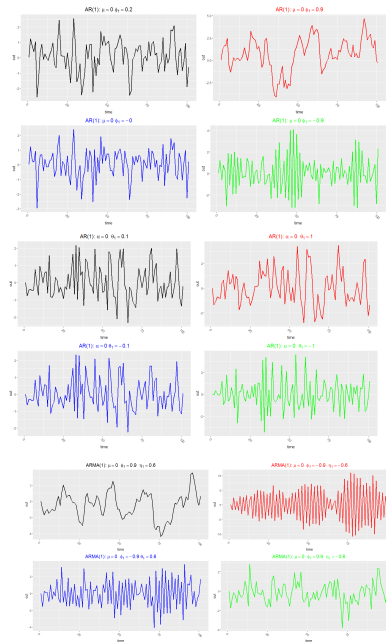
$$\text{AR}(1): Y_t = \mu + \phi_1 Y_{t-1} + \varepsilon_t \quad \varepsilon_t \sim WN(\sigma^2)$$

Moving Average : weighted average of recent 'shocks'

$$\text{MA}(1): Y_t = \alpha + \varepsilon_t + \theta_1 \varepsilon_{t-1} \quad \varepsilon_t \sim WN(\sigma^2)$$

Autoregressive Moving Average: AR(1) & MA(1) Component

$$\text{ARMA}(1,1): Y_t = \mu + \phi_1 Y_{t-1} + \varepsilon_t + \eta_1 \varepsilon_{t-1} \quad \varepsilon_t \sim WN(\sigma^2)$$

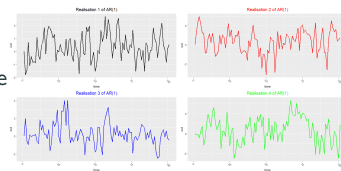


Fundamental Aim of Time Series

Suppose true unobservable process is $Y_t = 0.1 + 0.4Y_{t-1} + \epsilon_t$ where $\epsilon_t \sim WN(\sigma^2)$

Since $\{y_t\}_{t=1}^T$ drawn from Y_t - *should reflect properties of the process (under some assumptions)*

How to correctly infer properties of Y_t **given sample realisation** $\{y_t\}_{t=1}^T$?



Difficult to tell from the plotting the graph alone

Require framework to succinctly describe properties of a process...

Auto-covariance & Autocorrelation Function

Definition: Autocovariance Function of Y_t

$$\gamma(k) := Cov(Y_t, Y_{t-k}) \quad k = 0, 1, 2, \dots$$

Definition: Autocorrelation Function of Y_t

$$\rho(k) := \frac{Cov(Y_t, Y_{t-k})}{\sqrt{Var(Y_t)Var(Y_{t-k})}} = \frac{\gamma(k)}{\gamma(0)} \quad k = 0, 1, 2, \dots$$

$\rho(k)$ for $k = 1, 2, \dots$ reflect dependence properties of Y_t

We can derive $\rho(k)$ for different forms of processes

Correlation in sample data should reflect the correlation in Y_t

Can use this idea to infer form of Y_t (Lecture 3 & 4)

Theoretical Properties of MA Process

MA(1) Process: Mean & Variance

$$\text{MA(1): } Y_t = \alpha + \varepsilon_t + \theta_1 \varepsilon_{t-1}, \quad \varepsilon_t \sim WN(\sigma^2)$$

Mean of MA(1)

$$\begin{aligned}\mathbb{E}[Y_t] &= \mathbb{E}[\alpha + \varepsilon_t + \theta_1 \varepsilon_{t-1}] \\ &= \alpha + \theta_1 \mathbb{E}[\varepsilon_{t-1}] + \mathbb{E}[\varepsilon_t] \\ &= \alpha\end{aligned}$$

Variance of MA(1)

$$\begin{aligned}\text{Var}[Y_t] &= \mathbb{E}[(Y_t - E[Y_t])^2] \\ &= \mathbb{E}[(Y_t - \alpha)^2] \\ &= \mathbb{E}[(\varepsilon_t + \theta_1 \varepsilon_{t-1})^2] \\ &= \mathbb{E}[\varepsilon_t^2] + \theta_1^2 \mathbb{E}[\varepsilon_{t-1}^2] + 2\theta_1 \mathbb{E}[\varepsilon_t \varepsilon_{t-1}] \\ &= \sigma^2(1 + \theta_1^2)\end{aligned}$$

MA(1) Process: Autocovariance Function

Autocovariance Function of MA(1)

$$\begin{aligned}\text{Cov}[Y_t, Y_{t-1}] &= \mathbb{E}[(Y_t - \mathbb{E}[Y_t])(Y_{t-1} - \mathbb{E}[Y_{t-1}])] \\&= \mathbb{E}[(Y_t - \alpha)(Y_{t-1} - \alpha)] \\&= \mathbb{E}[(\varepsilon_t + \theta_1 \varepsilon_{t-1})(\varepsilon_{t-1} + \theta_1 \varepsilon_{t-2})] \\&= \mathbb{E}[\varepsilon_t \varepsilon_{t-1} + \theta_1 \varepsilon_t \varepsilon_{t-2} + \theta_1 \varepsilon_{t-1}^2 + \theta_1^2 \varepsilon_{t-1} \varepsilon_{t-2}] \\&= \mathbb{E}[\varepsilon_t \varepsilon_{t-1}] + \theta_1 \mathbb{E}[\varepsilon_t \varepsilon_{t-2}] + \theta_1 \mathbb{E}[\varepsilon_{t-1}^2] + \theta_1^2 \mathbb{E}[\varepsilon_{t-1} \varepsilon_{t-2}] \\&= \theta_1 \mathbb{E}[\varepsilon_{t-1}^2] \\&= \theta_1 \sigma^2\end{aligned}$$

$$\begin{aligned}\text{Cov}(Y_t, Y_{t-2}) &= \mathbb{E}[(\varepsilon_t + \theta_1 \varepsilon_{t-1})(\varepsilon_{t-2} + \theta_1 \varepsilon_{t-3})] \\&= 0\end{aligned}$$

$$\text{Cov}(Y_t, Y_{t-j}) = 0 \quad j \geq 2$$

MA(q) Process

Definition: Moving Average Process of Order q (MA(q))

$$Y_t = \alpha + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q} \quad \varepsilon_t \sim \text{WN}(\sigma^2)$$

for some coefficients $\alpha, \theta_1, \dots, \theta_q$ where $\theta_0 = 1$ (normalisation).

► **Mean of MA(q):** $\mathbb{E}[Y_t] = \alpha$

► **Variance of MA(q):** $\text{Var}[Y_t] = \sigma^2 (1 + \theta_1^2 + \dots + \theta_q^2)$

► **Autocovariance Function of MA(q):** $\gamma(k) = \begin{cases} \sigma^2 \sum_{s=0}^{q-k} \theta_s \theta_{s+k} & : k = 0, \dots, q \\ 0 & : k > q \end{cases}$

Recap

Lecture Recap

Distinction between true (unobserved) **process** Y_t and sample **realisation** (data) y_t

Plotted realisation of some time series variables

Key Definitions & Concepts- **Stationarity & Autocovariance/correlation Functions**

Introduced Simple Univariate Time Series Processes (**MA, AR, ARMA**)

Theoretical **Properties of MA(q)** Models (mean, variance and ACFs)

Video clips available in these slides, lectures notes & **Youtube Channel**

Next Week

Introduce $MA(\infty)$ and discuss its importance

Properties of $AR(1)$, $ARMA(1,1)$ and introduce general $ARMA(p,q)$ process

In particular discuss Autocorrelation/covariance Function of $AR(1)$, $ARMA(1,1)$

Conditions for stationarity in ARMA models

Reading: **Chapter 1** (this weeks material) and read **Chapter 2** in preparation for next week.